

RESEARCH ARTICLE

A Novel Contactless Approach to Continuous Blood Pressure Monitoring Using Complex CEEMDAN and Neural Networks

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ABSTRACT Contactless blood pressure (BP) monitoring is essential for daily healthcare. Typically, existing methods based on radar or remote photoplethysmography (rPPG) estimate only systolic blood pressure (SBP) and diastolic blood pressure (DBP). However, continuous monitoring of the complete blood pressure waveform provides deeper insights into cardiovascular health. This paper proposes a novel contactless continuous blood pressure monitoring algorithm termed CCNN-BP (Complex CEEMDAN and neural networks for blood pressure monitoring). Radar signals reflected from the chest are decomposed using Complex CEEMDAN (complete ensemble empirical mode decomposition with adaptive noise), to extract heartbeat-related intrinsic mode functions (IMFs). These IMFs are fed into a specialized neural network trained to estimate continuous blood pressure waveforms with accurate morphology. The final blood pressure waveform is obtained by adjusting the amplitude through calibration factors in combination with waveform characteristics. The proposed algorithm was trained and validated using an open-source dataset from Hamburg University of Technology. Experimental results show that the method achieves a mean absolute error (MAE) of 4.79 mmHg and a root mean square error (RMSE) of 7.13 mmHg for continuous blood pressure amplitude estimation, with a Temporal Pearson correlation coefficient of 0.86. Furthermore, the mean normalized dynamic time warping (DTW) distance was 0.94 ± 0.54 mmHg, confirming high morphological similarity. For peak delay evaluation, the method attains a MAE of 25.98 ms and RMSE of 34.79 ms, while achieving a Spectral Pearson correlation coefficient of 0.94, indicating robust preservation of key frequency components. These results demonstrate the feasibility of CCNN-BP for accurate, continuous, and contactless blood pressure waveform monitoring.

INDEX TERMS Blood pressure waveform, complex CEEMDAN, contactless sensing, continuous monitoring, neural network, radar signal processing.

I. INTRODUCTION

Blood pressure (BP) monitoring is crucial for the early diagnosis and effective management of cardiovascular diseases, particularly hypertension, a leading cause of morbidity worldwide. Conventional blood pressure monitoring methods, such as sphygmomanometers or oscillometric devices, are predominantly cuff-based systems that intermittently measure systolic blood pressure (SBP) and diastolic blood

pressure (DBP). Despite their accuracy, these traditional techniques require physical contact, causing discomfort during prolonged or frequent usage, and thus are unsuitable for continuous monitoring in everyday scenarios [1].

To enable continuous and noninvasive BP monitoring, researchers have explored techniques based on photoplethysmography (PPG), which detects volumetric changes in blood flow using optical sensors. PPG has become widely adopted in wearable devices due to its simplicity, low cost, and ability to capture cardiac-related information. Advanced models such as DeepCNAP [2] have demonstrated that deep

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TABLE 1. Summary of non-cuff blood pressure estimation methods.

Category	Measurement Principle	Sensor Type	Measurement Site	Output Type	Representative Studies
Contact PPG	Light absorption due to blood volume changes	PPG sensor (LED + photodiode)	Finger, wrist, earlobe	SBP/DBP/Waveform	Kim et al., 2022 [2], etc.
Remote PPG (rPPG)	Skin color or surface motion variation	RGB or high-speed camera	Face (cheek, forehead)	SBP/DBP	Jeong et al., 2016 [3], Schrumpt et al., 2021 [4], etc.
Radar-based	Micro-motion detection from chest wall	CW or FMCW radar	Chest, thorax	SBP/DBP	Wang et al., 2025 [5], Singh et al., 2023 [6], Zhao et al., 2018 [7], Zhao et al., 2024 [8], etc.

learning can reconstruct full arterial BP waveforms from contact-based PPG signals. However, the need for direct skin contact remains a limitation in scenarios requiring comfort, hygiene, or unobtrusiveness.

To overcome this, researchers have extended PPG into remote photoplethysmography (rPPG), which extracts physiological signals by analyzing subtle skin color changes captured via RGB cameras [3], [4]. rPPG methods eliminate the need for physical contact, but they are highly sensitive to motion artifacts and ambient lighting conditions, and may raise privacy concerns due to the use of facial video data.

In parallel, radar-based approaches have emerged as another class of fully contactless solutions. By analyzing chest wall micro-motions induced by cardiovascular activity, continuous-wave (CW) and frequency-modulated continuous-wave (FMCW) radar systems can detect heartbeat-related signals without relying on ambient light or visual imaging [5], [6], [7], [8]. These systems offer enhanced privacy and robustness, making them attractive for real-world applications.

Although these three approaches (PPG, rPPG and radar) can all track cardiac-related signals, there are substantial differences in their physiological signal sources, measurement principles and sensor placements. PPG typically uses green or infrared light and a photodetector placed on the finger, wrist, or earlobe to detect blood volume changes in capillaries. rPPG captures similar signals from the facial region, analyzing color fluctuations in skin pixels. In contrast, radar-based systems operate by transmitting microwaves toward the chest or carotid region, detecting sub-millimeter surface displacements caused by arterial pulsations. Unlike PPG and rPPG, radar is light-independent, non-intrusive, and capable of measuring through clothing or bedding.

Table 1 lists representative methods for estimating non-cuff blood pressure, including contact PPG, rPPG and radar-based approaches. For each category, the typical types of sensor, the underlying principles and the output characteristics are summarized. Due to space constraints, only a subset of relevant studies is included.

Although promising, current radar-based and rPPG methods largely focus on discrete BP measurements (SBP and

DBP), lacking the capability to reconstruct continuous blood pressure waveforms in a contactless manner. This limitation restricts their applicability in scenarios that demand high-resolution, real-time cardiovascular insights—such as peri-operative care, ICU monitoring, or at-home dynamic risk assessment.

In contrast, continuous blood pressure waveforms provide beat-to-beat representations of arterial pressure, allowing for much deeper clinical insight. These waveforms enable the analysis of key physiological parameters, including vascular compliance, cardiac output, pulse wave velocity, baroreflex sensitivity, and autonomic nervous system activity. They are also essential for capturing transient or subtle cardiovascular changes (e.g., arrhythmias, vasovagal events, or postural variations) that discrete SBP/DBP readings may overlook. In clinical analogs such as the CNAP system, waveform-level analysis has become a cornerstone for real-time, noninvasive hemodynamic monitoring.

However, realizing continuous waveform estimation using contactless techniques remains a challenging task due to the nonstationary, nonlinear nature of radar signals, and the difficulty in accurately isolating and interpreting heartbeat-related components.

To address this challenge, we propose CCNN-BP, a novel contactless algorithm for continuous blood pressure waveform monitoring. CCNN-BP processes radar signals reflected from the thoracic region—rich in physiological micro-motions—by first applying Complex CEEMDAN (complete ensemble empirical mode decomposition with adaptive noise) to extract intrinsic mode functions (IMFs) related to heartbeat-induced vibrations. These IMFs are then fed into a specialized hybrid neural network composed of a 1D convolutional U-Net, BiLSTM, and Transformer encoder, which is trained to reconstruct the morphologically accurate arterial pressure waveform. To correct amplitude distortions, a subject-specific calibration module based on pulse transit time (PTT) and empirically derived coefficients is employed.

The CCNN-BP algorithm was trained and validated using publicly available radar and CNAP data from a dataset provided by Hamburg University of Technology.

The remainder of this paper is organized as follows: Section II describes the CCNN-BP algorithm in detail. Section III introduces the open-source dataset used in this study and details the data preprocessing steps. Section IV presents the experimental results, including performance analysis. Finally, Section V provides concluding remarks and discusses future directions.

II. PROPOSED CCNN-BP ALGORITHM

A. SIGNAL MODEL

Radar-based monitoring of human vital signs relies on accurately measuring subtle surface displacements of the human body caused by physiological activities, such as respiration and heartbeat. These displacements are captured

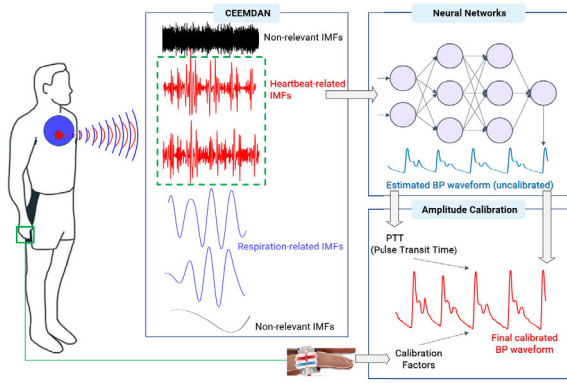


FIGURE 1. The overall framework of the proposed CCNN-BP algorithm. Radar signals are decomposed using Complex CEEMDAN to extract heart-beat-related IMFs, which are input into neural networks to estimate continuous blood pressure waveforms. Final waveform amplitudes are calibrated individually using two components: PTT derived from waveform characteristics, and personalized calibration factors.

as variations in the distance between the radar and the thoracic surface. In radar systems, such distance variations are encoded within the phase of the received complex radar baseband signal. To obtain the complex baseband signal used in our analysis, the radar receiver employs quadrature demodulation, which separates the received electromagnetic wave into two orthogonal components: the In-phase (I) component and the Quadrature (Q) component. These are obtained by mixing the signal with cosine and sine versions of the carrier signal, respectively. Together, they form a complex signal of the form

$$s(t) = I(t) + jQ(t) \quad (1)$$

which preserves both amplitude and phase information. This complex representation is essential for accurately capturing the subtle surface displacements of the chest wall due to physiological activity. By analyzing phase changes over time, vital sign-related motions can be effectively quantified.

Typically, the thoracic displacement induced by human vital signs can be modeled as:

$$R(t) = R_0 + R_b(t) + R_h(t) \quad (2)$$

where R_0 represents the static reference distance between radar and thoracic surface, $R_b(t)$ denotes displacement due to respiratory motion, and $R_h(t)$ denotes displacement due to heartbeat motion. The amplitude of respiratory displacement $R_b(t)$ generally falls within the order of centimeters, whereas heartbeat displacement $R_h(t)$ is typically in the millimeter range.

Assuming thoracic vibrations caused by respiration and heartbeat occur approximately in the same region and their amplitudes are linearly superposed spatially, the radar baseband received signal in a single-point scattering model can be expressed as:

$$s(t) = |X_0| e^{j\frac{4\pi}{\lambda}(R_b(t)+R_h(t))} = |X_0| e^{j\phi_{1p}(t)} \quad (3)$$

where $|X_0|$ represents the effective received signal magnitude of the target, λ is the carrier wavelength, and $\phi_{1p}(t)$ denotes the phase term of the single-point scattering model. Due to its simplicity, this single-point scattering model is widely adopted for initial vital sign extraction.

After extracting the phase $\phi_{1p}(t)$ from the radar signal, it is converted into the corresponding displacement $R(t)$. The resulting displacement signal $R(t)$ contains both respiratory motion $R_b(t)$ and cardiac motion $R_h(t)$. To isolate the heartbeat-related displacement, filtering techniques are typically applied to separate the higher-frequency cardiac component from the low-frequency respiratory component.

However, considering the physiological characteristics of thoracic motion, respiratory motion predominantly causes large-scale thoracic fluctuations, whereas heartbeat motion results in much smaller localized thoracic vibrations. Due to the significantly different distribution of these two motions on the thoracic surface, the scattering cross-sections associated with respiratory and cardiac activities differ considerably, leading to distinct amplitudes in their complex radar echo signals. Therefore, a more accurate approach is the two-point scattering model, proposed by Czerkawski et al. [9], which treats respiratory and heartbeat motions as originating from separate scattering points:

$$s(t) = |X_r| e^{j\frac{4\pi R_b(t)}{\lambda}} + |X_h| e^{j\frac{4\pi R_h(t)}{\lambda}} = |X'_0| e^{j\phi_{2p}(t)} \quad (4)$$

where $|X_r|$ and $|X_h|$ represent the effective magnitudes of the received signals associated with respiratory and heartbeat motions, respectively. $|X'_0|$ denotes the magnitude of the equivalent received signal after the superposition of the two independent scattering sources. The resultant phase term $\phi_{2p}(t)$ for the two-point model can be expressed as:

$$\phi_{2p}(t) = \arctan \left(\frac{|X_r| \sin\left(\frac{4\pi}{\lambda} R_b(t)\right) + |X_h| \sin\left(\frac{4\pi}{\lambda} R_h(t)\right)}{|X_r| \cos\left(\frac{4\pi}{\lambda} R_b(t)\right) + |X_h| \cos\left(\frac{4\pi}{\lambda} R_h(t)\right)} \right) \quad (5)$$

Given that respiratory motion not only has a larger amplitude but also affects a broader region compared to heartbeat motion, $|X_r| \gg |X_h|$. Under this condition, the phase term $\phi_{2p}(t)$ approaches $\frac{4\pi}{\lambda} R_b(t)$. Consequently, extracting accurate heartbeat information from the single-point scattering assumption can result in severe distortions. The two-point scattering model thus provides a more comprehensive representation of thoracic dynamics and serves as a robust foundation for precise extraction of both respiration and heartbeat components necessary for continuous, contactless blood pressure monitoring.

B. COMPLEX CEEMDAN DECOMPOSITION

In traditional heartbeat signal extraction methods based on the single-point scattering model, the standard processing pipeline typically includes:

- 1) Utilizing an ellipse fitting algorithm on the complex radar signal to correct offset, gain, and phase errors present in the sampled data [10];
- 2) Extracting the distance information from the corrected complex signal's phase;
- 3) Filtering the distance information to obtain the fluctuation waveform corresponding to the heartbeat signal [11].

However, since this study abandons the single-point scattering model due to its limitations, the conventional approach is no longer applicable. Instead, we adopt Complex CEEMDAN, an advanced variant of EMD [12], [13], to directly separate the heartbeat signal from the complex radar signal. This extracted heartbeat-related signal can subsequently be directly fed into the neural network, as described in the next section.

The CEEMDAN method is particularly suitable for decomposing intricate, nonlinear, and non-stationary signals frequently encountered in biomedical signal processing. By applying Complex CEEMDAN decomposition to the complex radar signal, we not only effectively isolate the heartbeat signal from respiratory motions and clutter but also simultaneously separate offset, gain, and phase errors inherently present in the sampled radar signal.

The decomposition of the radar signal $s(t)$ using Complex CEEMDAN can be expressed as:

$$s(t) = \sum_{k=1}^K \widetilde{IMF}_k(t) + \sum_{l=1}^L \widetilde{IMF}_l(t) + Res(t) \quad (6)$$

where:

- $\sum_{l=1}^L \widetilde{IMF}_l(t)$ represents the IMFs directly related to heartbeat motion;
- $\sum_{k=1}^K \widetilde{IMF}_k(t)$ corresponds to IMFs unrelated to heartbeat activity, such as respiratory signals, clutter, and noise components;
- $Res(t)$ denotes the residual component remaining after decomposition.

Finally, the complex IMFs corresponding to heartbeat-related signals, defined as:

$$s_h(t) = \sum_{l=1}^L \widetilde{IMF}_l(t) \quad (7)$$

are subsequently provided as input to the neural network detailed in the following section, facilitating accurate and robust estimation of blood pressure waveforms.

C. NEURAL NETWORK FOR BLOOD PRESSURE ESTIMATION

The input to the neural network consists of the complex IMFs corresponding to heartbeat-related signals introduced in the preceding subsection. The proposed neural network architecture for continuous blood pressure estimation consists of three main modules, as shown in Fig. 2: a one-dimensional

convolutional U-Net, a bidirectional long short-term memory (BiLSTM) network, and a Transformer encoder.

The first module employs a U-Net-inspired architecture tailored specifically for one-dimensional complex radar signals. Its primary objective is feature extraction through an encoder-decoder structure that captures multi-scale temporal features. The convolutional encoder progressively reduces the temporal resolution of the input signal while increasing feature channels, effectively compressing signal information. Correspondingly, the decoder reconstructs higher temporal resolution outputs, incorporating additive skip connections from the encoder layers to preserve essential fine-grained details crucial for accurate waveform morphology. Each block in the encoder and decoder utilizes convolutional layers followed by batch normalization and Leaky ReLU activations to ensure stable and efficient training. This structure additionally integrates dropout layers to mitigate overfitting.

After convolutional feature extraction, a BiLSTM module is employed to capture complex temporal dependencies within the extracted features. This dual-directional approach allows the network to accurately analyze temporal patterns intrinsic to physiological signals, such as cyclic heartbeats and subtle hemodynamic variations, significantly enhancing the network's capability to estimate precise blood pressure waveforms.

To further enhance the ability to model intricate temporal relationships within the sequence data, a Transformer encoder is integrated as the final analytical module. Transformer encoders leverage self-attention mechanisms, effectively capturing complex global interactions across different temporal points without positional constraints, unlike traditional recurrent models. The Transformer encoder adopted in this work comprises multi-head attention modules with positional encoding added to the input, followed by feedforward neural networks and normalization layers. This configuration ensures that the network fully exploits the temporal and contextual dependencies within radar-derived physiological signals.

Finally, the integrated outputs from the Transformer encoder are processed through linear transformation and Leaky ReLU activation to yield a one-dimensional output representing continuous arterial pressure waveforms with correct morphology. This structure additionally integrates dropout layers in each module to mitigate overfitting. This waveform is initially uncalibrated. Amplitude calibration is then performed, which is introduced in the next section. Fig. 2 also shows the parameter settings for the neural network.

A special loss function for neural network training has been developed for continuous arterial pressure waveforms, consisting of two main components:

- 1) Shape-aware Mean Squared Error (S-MSE):

This loss includes two sub-components with equal weight (50% each), aiming to capture both the morphological features and numerical accuracy of blood pressure waveforms.

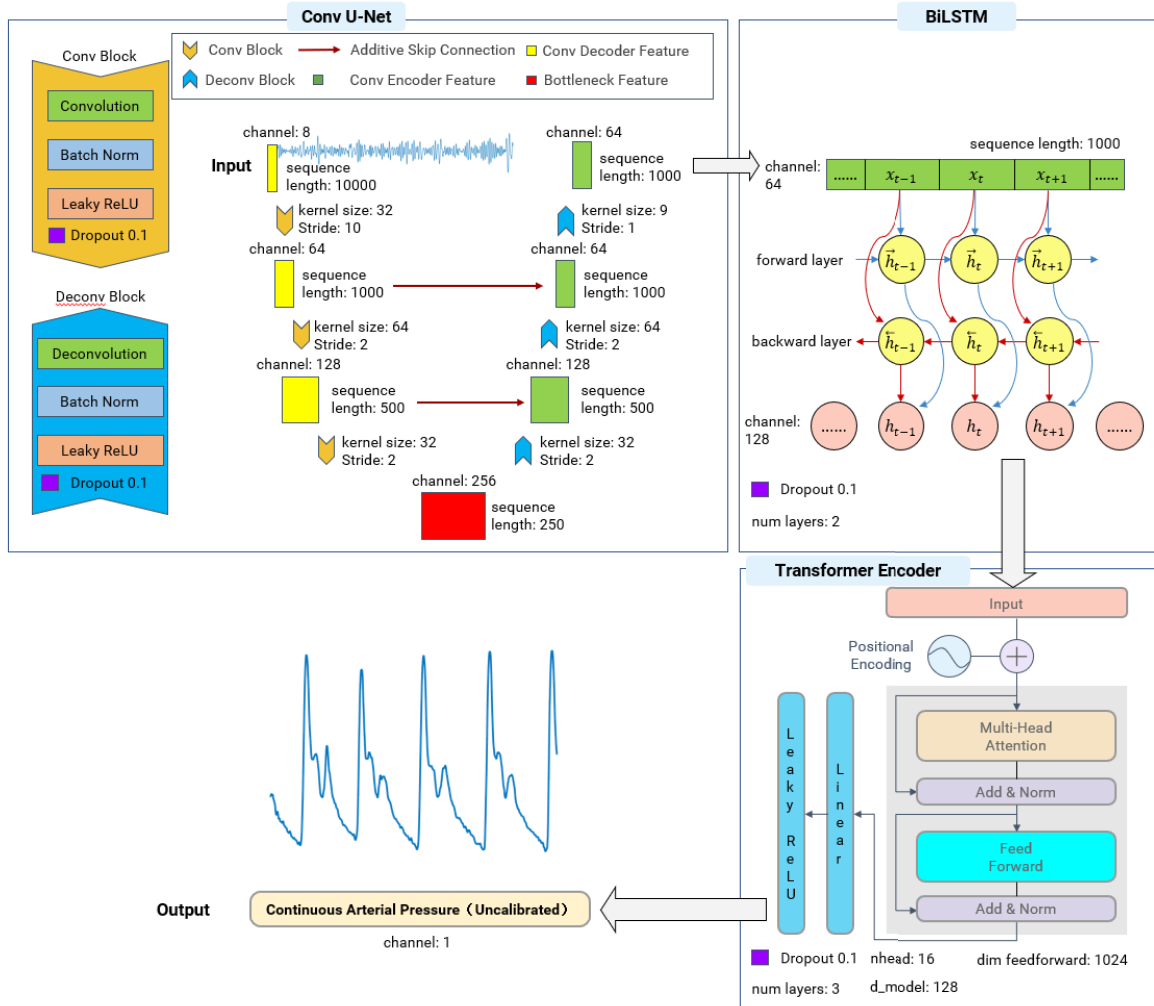


FIGURE 2. Schematic Diagram of the Proposed Neural Network Architecture. This architecture comprises a one-dimensional convolutional U-Net for feature extraction, a BiLSTM network for capturing temporal dependencies, and a Transformer encoder to model intricate global temporal interactions, generating uncalibrated continuous arterial pressure waveforms.

- Shape Loss: Captures morphological similarity. Defined as:

$$L_{\text{shape}} = \alpha \cdot \frac{1}{N} \sum_{i=1}^N (\hat{x}_i - x_i)^2 + \beta \cdot \frac{1}{N-1} \sum_{i=1}^{N-1} (\nabla \hat{x}_i - \nabla x_i)^2 + \gamma \cdot \frac{1}{N-2} \sum_{i=1}^{N-2} (\nabla^2 \hat{x}_i - \nabla^2 x_i)^2 \quad (8)$$

where $\hat{x}$ and x represent normalized predicted and target waveforms, ∇ denotes the first-order derivative (slope), and ∇^2 represents the second-order derivative (curvature). The hyperparameters α , β , and γ control the relative importance of these terms. This loss function evaluates not only point-wise differences but also how similar the waveform's overall shape is. The first-order term compares the trend of

rising and falling, and the second-order term helps preserve physiological structures like the diastolic notch.

Note that the denominators N , $N-1$, and $N-2$ are used because each derivative computation reduces the number of valid values:

- The amplitude term uses all N values;
- The slope (first derivative) yields $N-1$ values;
- The curvature (second derivative) yields $N-2$ values.

This normalization ensures each component contributes fairly to the overall loss.

- Mean Squared Error (MSE): Measures the direct discrepancy between predicted and target waveforms without normalization:

$$L_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 \quad (9)$$

where $\hat{y}$ and y represent the unnormalized predicted and target waveforms, respectively. This is a standard loss used in regression tasks and directly penalizes absolute amplitude errors.

While the shape loss emphasizes morphological similarity, the MSE loss ensures that the predicted waveform aligns with the target in real-world blood pressure values. These two losses together encourage the model to maintain physiological shape and clinical precision.

2) L2 Regularization (L2 Loss):

This component introduces a penalty term proportional to the squared magnitude of model parameters:

$$L_{L2} = \eta \cdot \sum_j w_j^2 \quad (10)$$

where η is a tunable coefficient controlling the regularization strength, and w_j represents each learnable parameter in the neural network. L2 regularization helps control the model complexity by discouraging excessively large weights. This is critical in deep networks, which often contain millions of parameters. Without this constraint, the model may memorize noise or training artifacts, resulting in poor generalization to new subjects.

Conceptually, L2 regularization acts like a soft force pulling the model toward simpler, smoother solutions. In biomedical signal modeling with limited data, it helps improve robustness and prevents overfitting.

Finally, we define the overall training objective as a weighted combination of the shape-aware loss (Equation 8), the mean squared error (Equation 9), and the L2 regularization term (Equation 10):

$$L_{total} = 0.5 \cdot (L_{shape} + L_{MSE}) + L_{L2} \quad (11)$$

This composite formulation is not taken from a specific reference, but follows standard practice in neural network-based time-series modeling. The first term equally combines waveform morphology (via L_{shape}) and amplitude accuracy (via L_{MSE}) with a 0.5 weighting factor each, ensuring the model learns both structural and quantitative aspects of blood pressure signals. The second term, L_{L2} , regularizes the network by penalizing large weights, which helps prevent overfitting and improves generalization to unseen subjects.

The choice of combining these terms reflects empirical findings from our validation experiments and aligns with common strategies in biomedical deep learning applications where both waveform shape and clinical accuracy are essential.

D. AMPLITUDE CALIBRATION

Although the neural network mentioned in the previous subsection successfully estimates continuous arterial pressure waveforms with accurate morphological alignment, discrepancies in amplitude can still arise due to limited

training data or inherent model uncertainties. SBP and DBP correspond respectively to the peaks and troughs of the continuous arterial pressure waveforms. To address these amplitude discrepancies, an amplitude calibration step is introduced to refine the estimated waveforms further.

The calibration process leverages PTT, a critical physiological parameter that inversely correlates with blood pressure. PTT is iteratively calculated from the estimated continuous arterial pressure waveform, employing the method introduced by Gao et al. [14]. Once PTT is accurately determined, SBP and DBP are recalibrated using empirically derived formulas proposed by Esmaili et al. [15]:

The calibration process leverages PTT, a critical physiological parameter that inversely correlates with blood pressure. In this study, the waveform estimated from radar reflections is treated as a surrogate for the radial arterial pressure waveform, denoted as $p_r(t)$, which represents the peripheral blood pressure signal typically observed at the radial artery. PTT is then calculated iteratively using the adaptive transfer function (ATF) method introduced by Gao et al. [14], which models the pressure wave transmission through a simplified arterial system described by two physiological parameters: wave travel time T_d and reflection coefficient Γ .

A parameter grid search is conducted over the ranges:

$$T_d \in [0, 250] \text{ ms (step size: 5 ms)}, \quad \Gamma \in [0, 1] \text{ (step size: 0.05)}$$

For each candidate pair (T_d, Γ) , a synthetic central blood pressure waveform $p_c(t)$, representing the pressure at the aortic root, is synthesized from the estimated radial waveform $p_r(t)$ using the following equation:

$$p_c(t) = \frac{1}{1 + \Gamma} p_r(t + T_d) + \frac{\Gamma}{1 + \Gamma} p_r(t - T_d) \quad (12)$$

This synthesized waveform is then filtered with a 100-tap low-pass FIR filter with a cutoff frequency of around 8 Hz. Within each cardiac cycle, the diastolic segment is estimated as:

$$D = 0.4 \left(1 - e^{-2 \cdot PL}\right) \quad (13)$$

where PL denotes the pulse length. In each diastolic interval, an exponential decay is fitted to the waveform, following the log-linear model:

$$\ln(p_c(t)) = -kt + b \quad (14)$$

The root-mean-square error (RMSE) of this fit is calculated as:

$$E_{rms} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\ln(p_c(t_i)) + kt_i - b)^2} \quad (15)$$

The parameter pair that minimizes the cumulative fitting error is selected as:

$$(T_d^*, \Gamma^*) = \arg \min_{T_d, \Gamma} E_{rms} \quad (16)$$

The optimal T_d^* is adopted as the subject-specific PTT. Finally, SBP and DBP are estimated using empirically derived calibration functions proposed by Esmaili et al. [15]:

$$SBP = a_0 + \sqrt{a_1 + a_2 \frac{1}{PTT^2}} \quad (17)$$

$$DBP = b_0 + \sqrt{b_1 + b_2 \frac{1}{PTT^2}} \quad (18)$$

In these equations, (a_0, a_1, a_2) and (b_0, b_1, b_2) represent subject-specific calibration coefficients initially obtained by fitting the above equations to the actual continuous arterial pressure waveforms from each individual. Following this calibration, the amplitude and baseline offset of the neural network output waveform are adjusted to match these calibrated SBP and DBP values, effectively aligning the estimated waveform amplitude with physiological reality.

This amplitude calibration step significantly enhances the accuracy of continuous blood pressure monitoring, ensuring that both waveform morphology and amplitude correspond closely to the true physiological signals, thereby improving the clinical applicability of the proposed CCNN-BP method.

E. STEP-BY-STEP IMPLEMENTATION OF THE CCNN-BP ALGORITHM

To provide a clear and consolidated overview, the complete process of the proposed CCNN-BP algorithm for estimating a continuous blood pressure waveform from a raw radar signal segment is summarized in the following steps:

- 1) **Signal Decomposition and Input Preparation:** The process starts with a segment of the raw complex radar signal, $s(t)$. The Complex CEEMDAN algorithm is applied to decompose $s(t)$ into a set of intrinsic mode functions (IMFs). Based on spectral analysis, the IMFs that are primarily related to heartbeat activity are identified, which serves as the input to the neural network. This step corresponds to the methodology described in Section II-B.
- 2) **Neural Network-based Waveform Estimation:** The extracted heartbeat IMFs, are fed into the pre-trained CCNN-BP neural network. The network, comprising a 1D convolutional U-Net, a BiLSTM, and a Transformer encoder, processes the input sequence and generates an uncalibrated continuous arterial pressure waveform, $p_{\text{uncal}}(t)$. This output waveform accurately reflects the morphological characteristics of the true blood pressure signal but lacks correct amplitude scaling (as detailed in Section II-C).
- 3) **Pulse Transit Time (PTT) Calculation:** The uncalibrated waveform, $p_{\text{uncal}}(t)$, is used to calculate the subject's Pulse Transit Time (PTT). The ATF method is employed, which involves an iterative grid search over a range of physiological parameters (wave travel time T_d and reflection coefficient Γ) to find the optimal pair

that minimizes the fitting error of the diastolic decay, as outlined in Equation 12-16 in Section II-D. The optimal T_d is adopted as the PTT.

- 4) **SBP and DBP Value Calibration:** With the PTT value determined, the calibrated Systolic Blood Pressure (SBP_{cal}) and Diastolic Blood Pressure (DBP_{cal}) are calculated. This is achieved by using the PTT in the empirically derived formulas (Equation 17 and 18) along with subject-specific coefficients that are obtained in a one-time, personalized calibration phase.
- 5) **Final Waveform Amplitude Adjustment:** The final step is to adjust the amplitude of the neural network's output waveform. A linear scaling and offset transformation is applied to $p_{\text{uncal}}(t)$ to align its minimum and maximum values with the calibrated DBP_{cal} and SBP_{cal} from the previous step. The final, calibrated continuous blood pressure waveform, $p_{\text{final}}(t)$, is thereby produced.

This structured procedure ensures that both the morphological fidelity and the clinical amplitude accuracy of the blood pressure waveform are achieved in a contactless manner.

III. DATASET AND PREPROCESSING

The dataset employed in this research was generously provided as open-source by Schellenberger et al. from Hamburg University of Technology. It consists of 24 hours of synchronized data recorded from 30 healthy subjects under predefined experimental protocols. The data collection involved a 24 GHz continuous wave radar system operating in the ISM band to capture chest displacement signals related to thoracic movements. The radar IQ signal sampling rate is 2000 Hz. Radar IQ signals were sampled at a rate of 2000 Hz, while CNAP signals were simultaneously recorded by a reference device, the Task Force Monitor (TFM), at a sampling rate of 200 Hz [16], [17].

The experimental protocol included several scenarios designed to induce different physiological responses. In the resting scenario, subjects lay still and breathed calmly to provide baseline physiological data. The valsalva scenario involved forceful exhalation against a closed glottis, triggering significant hemodynamic changes. In the apnea scenario, subjects held their breath in two conditions: after full inhalation and after full exhalation, allowing the study of blood pressure changes during breath-holding. The tilt up and tilt down scenarios involved tilting the subjects on a tilt table to 70 degrees and returning them to a horizontal position, respectively, to stimulate the autonomic nervous system through postural changes.

Considering the extensive length of the original recordings, the data were segmented into 5-second intervals with a 30% overlap. The first 15 subjects were allocated to the training and validation datasets (split at 80% and 20%, respectively), while the data from the last 15 subjects constituted the testing dataset. For calibration purposes, 20 segments of data from

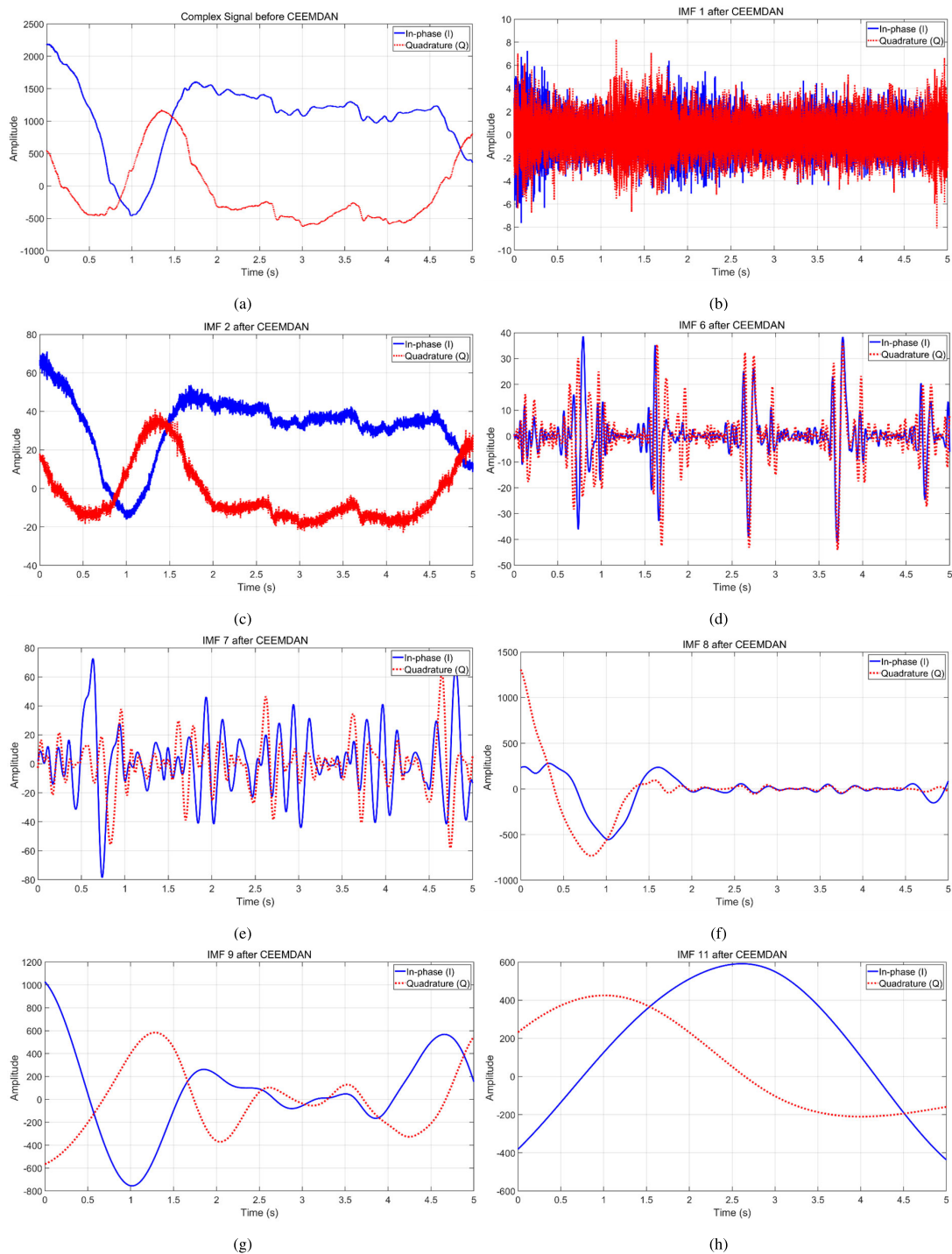


FIGURE 3. Complex CEEMDAN decomposition of radar signals (ID=5, Resting scenario). (a) Original complex radar IQ signal. The I (In-phase) and Q (Quadrature) components represent the real and imaginary parts of the radar baseband signal, respectively. They are derived via quadrature demodulation of the received electromagnetic wave and encode the amplitude and phase variations caused by physiological micro-motions; (b–c) Representative noise IMFs (IMF 1 and IMF 2); (d–e) Heartbeat-related IMFs (~68–84 bpm, IMF 6 and IMF 7); (f–g) Lower-frequency IMFs (~14–22 bpm, IMF 8 and IMF 9); (h) Signal offset component (IMF 11).

each subject in the testing dataset were individually used for calibration.

Detailed information regarding the subjects is provided in Table 2.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. CEEMDAN DECOMPOSITION ANALYSIS

Fig. 3 shows the decomposition results of a representative radar signal segment (ID=5, scenario: resting) using

TABLE 2. Subject information for model development.

ID	Age	Sex	Height (cm)	Weight (kg)	BMI
GDN0001	24	F	166	63	22.9
GDN0002	38	F	161	49	18.7
GDN0003	25	M	187	82	23.4
GDN0004	28	F	178	59	18.6
GDN0005	27	F	173	93	31.1
GDN0006	49	F	172	63	21.3
GDN0007	24	F	187	80	22.9
GDN0008	24	M	182	77	23.2
GDN0009	40	M	184	74	21.9
GDN0010	24	M	186	78	22.5
GDN0011	21	F	165	55	20.2
GDN0012	24	M	193	86	23.1
GDN0013	49	F	173	62	20.7
GDN0014	35	F	153	44	18.8
GDN0015	27	M	182	78	23.5
GDN0016	42	F	165	61	22.4
GDN0017	49	F	167	85	30.5
GDN0018	28	M	165	57	20.9
GDN0019	27	F	166	59	21.4
GDN0020	23	F	172	68	23.0
GDN0021	24	M	187	85	24.3
GDN0022	61	F	178	90	28.4
GDN0023	27	M	186	69	19.9
GDN0024	21	F	165	65	23.9
GDN0025	26	M	183	82	24.5
GDN0026	31	F	160	51	19.7
GDN0027	24	M	187	83	23.7
GDN0028	29	M	190	94	26.0
GDN0029	25	M	186	82	23.7
GDN0030	26	M	172	93	31.4

Complex CEEMDAN to demonstrate the extraction of heartbeat related IMFs. The decomposition yields a total of 12 IMFs, including the residual; however, due to space constraints, only a selection of key IMFs is displayed and analyzed. Fig. 3(a) shows the original complex radar signal (IQ signal), capturing the raw reflections from the subject's chest. IMF 1 (Fig. 3(b)) and IMF 2 (Fig. 3(c)) represent typical noise components in the original radar signals. These two IMFs capture the high-frequency noise that often appears prominently in initial IMFs derived from CEEMDAN decomposition. Additional noise-related IMFs are omitted due to brevity.

Heartbeat-related components were identified primarily in IMF 6 (Fig. 3(d)) and IMF 7 (Fig. 3(e)). These IMFs display periodic waveforms closely correlated with cardiac activity, with estimated periodicities corresponding roughly to a heart rate of 68–84 beats per minute (bpm).

Lower-frequency physiological signals associated with respiration or slower cardiac-related modulations appear distinctly in IMF 8 (Fig. 3(f)) and IMF 9 (Fig. 3(g)), showing periodicities corresponding to approximately 14–22 bpm.

Fig. 3(h) illustrates IMF 11, representing a low-frequency offset typically observed in radar-based measurements due to baseline shifts or signal drift throughout the recording duration.

Other IMFs, though part of the decomposition, are not presented due to space limitations. These selected IMFs demonstrate how Complex CEEMDAN effectively isolates critical signal components necessary for subsequent continuous blood pressure waveform estimation.

Moreover, it is evident from the comparison that the amplitude of the IQ signals in IMF 6 and IMF 7, which are associated with heartbeat activity, is significantly smaller than that observed in IMF 8 and IMF 9, which correspond to respiratory motion. This observation aligns well with the previously introduced two-point scattering model, further validating its underlying assumption. Specifically, the scattering surface induced by cardiac motion is considerably smaller than that caused by respiratory movement, leading to a correspondingly weaker radar reflection amplitude. Such a difference in scattering magnitudes reinforces the necessity of adopting a more refined scattering model for accurate physiological signal extraction in radar-based monitoring systems. Consequently, the application of Complex CEEMDAN decomposition, capable of separating these overlapping but amplitude-distinct components, is not only justified but essential for ensuring accurate isolation of heartbeat-related signals.

To provide further insight into the IMFs decomposed in Figure 3, we analyzed each component in terms of its spectral centroid and effective mean amplitude. These metrics help distinguish between noise, respiratory motion, baseline drift, and heartbeat-related components. Notably, the spectral centroid reflects the weighted average frequency of an IMF, while the effective mean amplitude emphasizes the signal's active segments by excluding near-zero regions. A complete summary of the 12 IMFs and the residual, along with their respective physiological interpretations, is presented in Appendix A.

B. ANALYSIS OF ESTIMATED BP WAVEFORMS

This section provides a detailed analysis of the estimated blood pressure waveforms generated by the proposed CCNN-BP algorithm, using publicly available radar IQ signal data synchronized with reference blood pressure waveforms collected through the TFM.

Fig. 4 presents the comparison between the estimated CCNN-BP waveforms and the TFM reference signals for Subject ID:23 across five different scenes. Overall, the estimated waveforms closely follow the reference signals, accurately capturing the primary morphological features, with minimal deviations observed around major peaks and valleys. Nevertheless, localized discrepancies are observed around the diastolic waves (small secondary peaks following the primary systolic peaks) that appear after the primary systolic peaks (the main highest points corresponding to ventricular ejection), where the alignment between the estimated and reference waveforms becomes less precise. These differences may arise from subtle variations in waveform morphology or measurement noise. Despite these localized deviations, the overall waveform tracking remains robust, demonstrating the model's ability to maintain general shape and timing fidelity across a range of physiological conditions.

To evaluate waveform morphological similarity while accounting for temporal misalignment, we employed dynamic time warping (DTW). We define the normalized

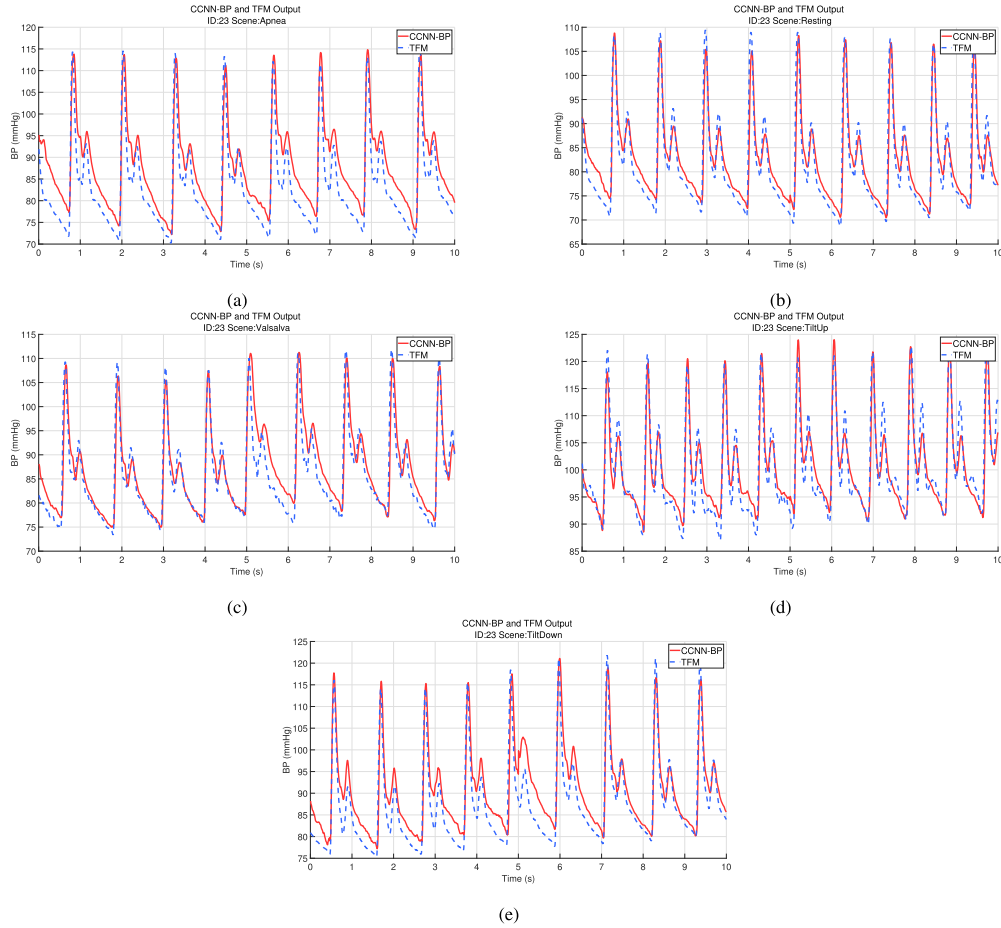


FIGURE 4. Comparison of estimated CCNN-BP and reference TFM blood pressure waveforms for Subject ID:23 across five scenes ((a): apnea, (b): resting, (c): valsalva, (d): tilt up, (e): tilt down). Red solid lines represent CCNN-BP estimations, and blue dashed lines represent TFM reference signals.

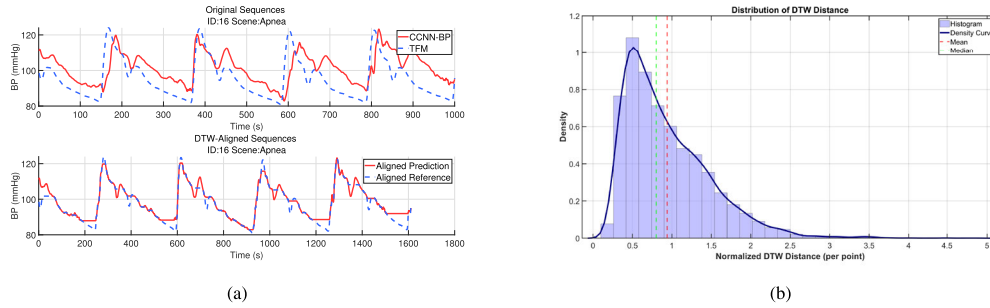


FIGURE 5. (a) Example of waveform alignment using DTW for sample ID: 16, Scene: Apnea. Although the raw predicted and reference waveforms are misaligned (RMSE = 10.24 mmHg, MAE = 8.94 mmHg, Temporal Pearson's $r = 0.66$), the normalized DTW distance is only 1.68 mmHg per point, indicating strong morphological similarity. This demonstrates the robustness of DTW to temporal lags; (b) Distribution of normalized DTW distances across all test waveform segments. Most values are concentrated around a low mean value, suggesting strong morphological alignment between predicted and reference waveforms.

DTW distance as the average squared alignment cost per matched point, calculated by dividing the total DTW distance by the length of the optimal alignment path:

$$\text{Normalized DTW} = \frac{1}{L} \sum_{(i,j) \in \text{path}} |x_i - y_j| \quad (19)$$

where L is the number of alignment steps. The resulting unit is mmHg, which reflects the average squared deviation between aligned waveform segments. This metric complements

traditional point-wise errors such as RMSE and MAE, especially under small phase lags. An example is illustrated in Fig. 5(a), where despite a visible time lag (reflected in a relatively high RMSE of 10.24 mmHg and a modest Pearson correlation of 0.66), the DTW alignment yields a normalized distance of only 1.68 mmHg per point. This highlights that the waveform shape is well preserved, and that the temporal misalignment significantly affects traditional point-wise metrics.

TABLE 3. Performance metrics of CCNN-BP algorithm Compared to TFM reference.

ME of Time Series (mmHg)	MAE of Time Series (mmHg)	RMSE of Time Series (mmHg)	Temporal Pearson's Correlation	Mean Normalized DTW distance (mmHg)
1.18	4.79	7.13	0.86	0.94
ME of Peak Delay (ms)	MAE of Peak Delay (ms)	RMSE of Peak Delay (ms)	Spectral Pearson's Correlation	Std. of Normalized DTW distance (mmHg)
14.17	25.98	34.79	0.94	0.54

TABLE 4. Comparison of SBP and DBP estimation performance with state-of-the-art radar-based methods.

Method	SBP (mmHg)				DBP (mmHg)			
	ME	MAE	SD	RMSE	ME	MAE	SD	RMSE
TRCCBP	–	4.73	5.52	–	–	4.49	5.25	–
RSD-Net	-0.32	4.86	6.14	6.14	-0.2	4.42	5.5	5.5
Proposed Method	1.08	4.91	6.97	7.06	-2	4.17	5.41	5.77

Table 3 summarizes various error metrics used to evaluate the performance of the proposed CCNN-BP algorithm. These metrics include the mean error (ME), mean absolute error (MAE), and root mean square error (RMSE) for both the estimated blood pressure waveform time series and the peak delay between waveforms. Additionally, two types of Pearson’s correlation coefficients are reported: the Temporal Pearson’s Correlation, which assesses the direct similarity between the estimated and reference time series, and the Spectral Pearson’s Correlation, which evaluates the similarity of the estimated and reference signals after spectral estimation. To further assess morphological similarity between waveforms in the presence of temporal shifts, the mean normalized DTW distance and its standard deviation are also reported. The normalized DTW distance is computed by dividing the total alignment cost (based on Euclidean distance) by the length of the optimal DTW path.

Spectral correlation analysis is particularly useful for evaluating the consistency of dominant frequency components between the estimated and reference signals. In continuous blood pressure waveform estimation, slight temporal shifts—such as small delays or advances in waveform peaks—can cause significant pointwise errors in time-domain correlation analysis, even when the overall morphological patterns are similar. These temporal misalignments can occur due to minor model inaccuracies, physiological variations, or measurement noise. By transforming the waveforms into the frequency domain and analyzing their spectral representations, the comparison becomes less sensitive to small temporal offsets, since frequency domain features primarily reflect the global oscillatory behavior rather than precise timing. As a result, spectral correlation provides a more robust evaluation of whether the algorithm successfully preserves the key oscillatory characteristics of blood pressure dynamics, even when slight temporal misalignments are present.

For the blood pressure waveform time series, the MAE and RMSE are 4.79 mmHg and 7.13 mmHg, respectively, with a Temporal Pearson’s correlation coefficient of 0.86, indicating

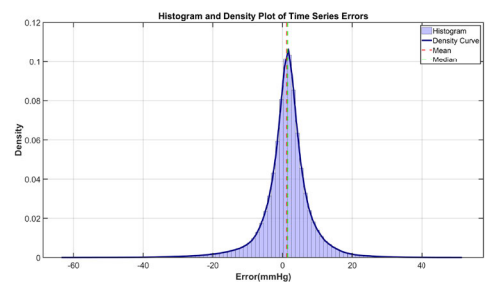


FIGURE 6. Histogram and density plot of time series errors between the CCNN-BP estimated blood pressure waveforms and the TFM reference. The blue curve represents the estimated probability density, with vertical dashed lines indicating the mean (red) and median (green) errors. The distribution is centered near zero, suggesting minimal systematic bias in amplitude estimation.

that the CCNN-BP algorithm can accurately reconstruct continuous blood pressure waveforms with high fidelity.

In terms of peak delay, the MAE and RMSE are 25.98 ms and 34.79 ms, respectively. The relatively low delay errors, considering the physiological transit time of blood from the chest to the fingertip (typically around 100–300 ms), demonstrate that the neural network effectively compensates for physiological delays. Furthermore, a high Spectral Pearson’s correlation coefficient of 0.94 confirms that the algorithm preserves the major frequency characteristics of the blood pressure waveform, even in the presence of minor temporal misalignments.

To further evaluate the performance of the proposed CCNN-BP algorithm, we compared it with two recently published radar-based contactless blood pressure estimation methods: TRCCBP (Transformer for Radar-based Contactless Continuous Blood Pressure, 2023) [18] and RSD-Net (Radar-based Stacked Deformable Convolutional Network, 2024) [19]. Both approaches use a radar sensor aimed at the chest to extract motion signals associated with cardiovascular activity, similar to our setup. Notably, RSD-Net was trained and validated using the same open-source dataset employed in this study, ensuring a fair basis for comparison.

Since our method generates continuous blood pressure waveforms, we extracted the systolic blood pressure (SBP) and diastolic blood pressure (DBP) values by locating the maximum and minimum points of each cardiac cycle to enable consistent evaluation against the discrete results provided by TRCCBP and RSD-Net.

Table 4 presents the comparison results in terms of ME, MAE, standard deviation (SD), and RMSE for both SBP and DBP. For transparency, note that the symbol “–” in the table indicates that the corresponding metrics were not reported in the original paper.

Although our method shows slightly higher values in some metrics, the differences are relatively small. More importantly, our method offers the advantage of estimating full continuous blood pressure waveforms, which provides richer physiological information and stronger clinical relevance compared to SBP/DBP-only outputs. This continuous waveform estimation supports more comprehensive cardiovascular

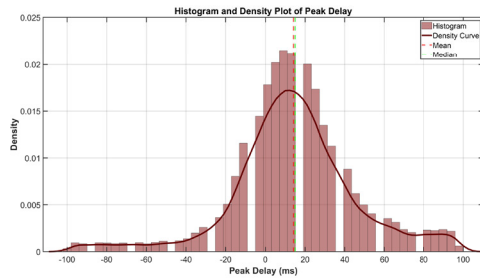


FIGURE 7. Histogram and density plot of peak delay between the CCNN-BP estimated waveforms and the TFM reference. The red curve illustrates the probability density distribution, and the vertical dashed lines denote the mean (red) and median (green) delays. Most delays are concentrated around small positive values, indicating effective but slightly lagged temporal alignment.

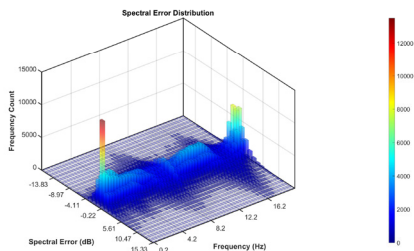


FIGURE 8. Spectral error distribution between the CCNN-BP estimated and TFM reference waveforms. The 3D histogram illustrates the frequency-wise distribution of spectral errors in dB. The concentration of low spectral errors at key physiological frequencies demonstrates the algorithm's capability to preserve major frequency components in blood pressure waveforms.

monitoring and potential integration into long-term health management systems.

Fig. 5(b) presents the distribution of normalized DTW distances computed over all test segments. The histogram reveals that the majority of DTW distances are tightly clustered around a low mean value, further confirming that the CCNN-BP algorithm produces morphologically accurate blood pressure waveforms. Despite occasional temporal misalignment, the waveform shape is consistently preserved.

To further evaluate the error distribution characteristics of the proposed CCNN-BP algorithm, histograms and density plots of time series errors and peak delay errors, as well as a three-dimensional spectral error distribution plot, are presented in Fig. 6, 7, and 8.

Fig. 6 shows the histogram and estimated probability density of the time series amplitude errors between the CCNN-BP estimated blood pressure waveforms and the TFM reference. The distribution is sharply peaked around zero, with the mean and median errors both located close to zero, indicating a small overall systematic bias. The near overlap between the mean and median lines suggests that the error distribution is approximately symmetric.

Fig. 7 illustrates the histogram and density curve of the peak delay errors. The majority of peak delays are distributed around small positive values, with a slight rightward skewness observed in the distribution. These results indicate that the CCNN-BP algorithm achieves effective

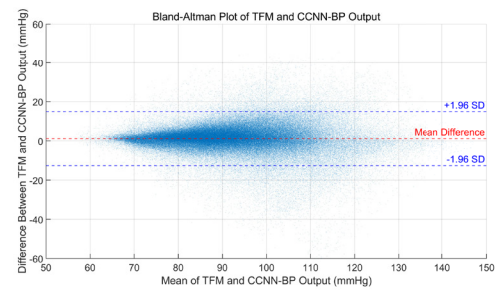


FIGURE 9. Bland-Altman plot comparing TFM and CCNN-BP outputs across all data points. The mean difference and limits of agreement (± 1.96 SD) were calculated using the entire dataset. The mean difference close to zero indicates minimal systematic bias, and the majority of points lie within the limits.

temporal alignment between the estimated and reference waveforms. Although a minor residual delay remains, the neural network successfully compensates for the inherent physiological delay associated with pulse wave propagation from the chest to peripheral sites.

Fig. 8 presents the spectral error distribution as a function of frequency. The three-dimensional histogram shows that spectral errors are generally low at key physiological frequencies, particularly within the heart rate (approximately 1 Hz) and respiratory rate (approximately 0.1–0.3 Hz) bands. This concentration of low spectral errors demonstrates the algorithm's ability to preserve the major oscillatory components of the blood pressure waveform, even in the presence of slight temporal misalignments that could affect time-domain evaluations.

Fig. 9 illustrates the Bland-Altman analysis between the outputs of TFM and CCNN-BP using all available data points across all volunteers. The mean difference is close to zero, suggesting negligible systematic bias between the two measurement methods. The limits of agreement, defined as the mean ± 1.96 standard deviations, encompass the majority of the data points, indicating good consistency and reliability. Although slight deviations occur, they are expected due to intrinsic waveform differences and minor temporal misalignments. Overall, the plot confirms a strong agreement between TFM and CCNN-BP outputs across the entire dataset.

In conclusion, despite certain limitations, these results affirm the potential of the CCNN-BP algorithm for accurate, continuous, and contactless blood pressure waveform monitoring.

C. ABLATION STUDY

To further evaluate the individual contributions of various components in the proposed CCNN-BP algorithm, we performed a detailed ablation study. This study aims to investigate the impact of the complex CEEMDAN decomposition and the roles played by each neural network submodule—namely, 1D convolutional U-Net (Conv), BiLSTM, and Transformer encoder (Transformer)—in accurate and robust continuous blood pressure waveform estimation.

TABLE 5. Ablation Study Results for CCNN-BP Components.

Metric	Model					
	No Complex CEEMDAN	Only Conv	BiLSTM + Transformer	Conv + BiLSTM	Conv + Transformer	Full CCNN-BP
ME of Time Series (mmHg)	2.09	4.08	2.45	1.62	2.10	1.18
MAE of Time Series (mmHg)	6.33	7.60	5.72	5.82	5.57	4.79
RMSE of Time Series (mmHg)	8.85	9.80	7.98	8.26	7.91	7.13
Temporal Pearson's Correlation	0.78	0.77	0.83	0.81	0.83	0.86
ME of Peak Delay (ms)	13.58	15.17	13.39	19.30	20.76	14.17
MAE of Peak Delay (ms)	29.93	26.73	34.80	36.23	34.34	25.98
RMSE of Peak Delay (ms)	38.49	36.49	43.09	44.33	43.68	34.79
Spectral Pearson's Correlation	0.88	0.88	0.87	0.88	0.88	0.94
Mean Normalized DTW distance (mmHg)	1.38	1.94	1.18	1.47	1.24	0.94
Std. of Normalized DTW distance (mmHg)	0.73	0.78	0.61	0.82	0.67	0.54

Six different model configurations were tested for comparison:

- (A) **No Complex CEEMDAN:** This configuration discards the proposed two-point scattering model and complex decomposition. Instead, a conventional approach is used—extracting chest displacement from phase signals using ellipse fitting, followed by real-valued CEEMDAN decomposition. The derived heartbeat-related IMFs are then fed into the same neural network. This baseline reflects the traditional single-point scattering signal model and assesses the necessity of the proposed radar signal modeling approach.
- (B) **Only Conv:** In this setting, the neural network includes only Conv module, while BiLSTM and Transformer modules are removed. It tests whether convolutional feature extraction alone is sufficient for temporal modeling.
- (C) **BiLSTM + Transformer (No Conv):** This variant removes Conv module, retaining BiLSTM and Transformer modules. It tests if temporal models alone can capture morphological features from CEEMDAN IMFs.
- (D) **Conv + BiLSTM (No Transformer):** This variant removes Transformer modules, retaining Conv and BiLSTM modules. This model evaluates the influence of excluding the Transformer encoder.
- (E) **Conv + Transformer (No BiLSTM):** This version omits BiLSTM and evaluates the Transformer's ability to handle temporal dynamics.
- (F) **Full CCNN-BP (Proposed):** The complete model with complex CEEMDAN input and full neural architecture (Conv + BiLSTM + Transformer) serves as the performance benchmark.

All configurations were trained and evaluated on the same dataset described in Section III, and identical post-processing and calibration procedures were applied.

Table 5 summarizes the performance of all model variants across multiple metrics, including ME, MAE, RMSE, temporal and spectral Pearson's correlation, and normalized DTW distance.

These results reveal several insights:

- Removing complex CEEMDAN decomposition (model A) leads to the most substantial decline in performance

across nearly all metrics, confirming the effectiveness of the two-point scattering model and complex signal separation.

- Using only Conv (model B) results in the worst MAE and RMSE, demonstrating that temporal modeling (BiLSTM and Transformer) is crucial for capturing inter-beat dynamics and waveform continuity.
- Retaining only BiLSTM and Transformer without Conv (model C) yields improved temporal correlation and DTW alignment compared to model B, highlighting the strong capacity of temporal models, although convolution still contributes to local feature extraction.
- Excluding either BiLSTM or Transformer (models D and E) results in the worst peak delay errors among all variants, indicating that both temporal modules are necessary for accurate timing estimation.
- The full CCNN-BP model (model F) consistently achieves the best overall performance in waveform accuracy, temporal alignment, and spectral fidelity, highlighting the complementary roles of all components.

To further illustrate how each component affects waveform fidelity, we selected a representative case (Subject ID:23, Scene: Resting) and plotted the estimated continuous blood pressure waveforms generated by each model configuration. Fig. 10 displays these waveform comparisons. Each subplot compares estimated BP waveform (red) and reference waveform from TFM (blue dashed line). The key differences are summarized below:

- (a) **No Complex CEEMDAN:** Without using complex CEEMDAN and the two-point scattering model, the estimated waveform exhibits significant loss in structural fidelity. Notably, a heartbeat peak around 5 seconds is entirely missed, indicating the algorithm's failure to preserve weak or interference-corrupted cardiac components. The waveform also displays reduced amplitude and morphological inconsistencies, particularly during the descending phases. These results confirm that complex decomposition is essential for isolating heartbeat-related IMFs and suppressing respiratory and motion artifacts embedded in the phase-derived signals.
- (b) **Only Conv:** Without BiLSTM and Transformer, the model shows obvious timing errors, especially between

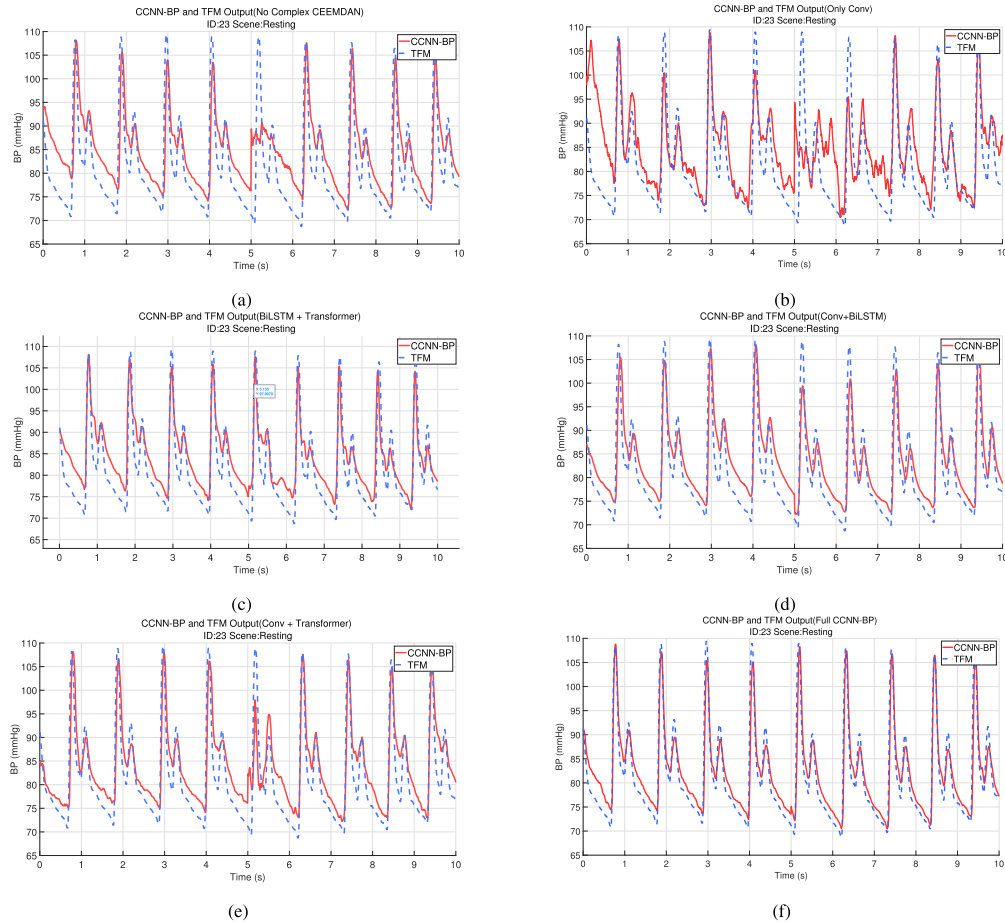


FIGURE 10. Ablation study visual results for Subject ID:23 (Resting). (a): No Complex CEEMDAN, (b): Only Conv, (c): BiLSTM + Transformer, (d): Conv+BiLSTM, (e): Conv + Transformer, (f) Full CCNN-BP.

5–7 seconds where several peaks are delayed or distorted. The waveform is noisy and unstable, and some key features like the dicotic notch are missing. This shows that convolution alone cannot capture the temporal structure of blood pressure signals.

- (c) **BiLSTM + Transformer:** Without convolution, the model maintains good timing and overall waveform shape. Most peaks are correctly located, and the rhythm is consistent. However, some fine details are smoothed out, and the waveforms appear less sharp, suggesting that convolution helps capture local morphological features.
- (d) **Conv + BiLSTM:** Removing the Transformer causes clear degradation in waveform shape. While the timing and rhythm are mostly preserved, many peaks are less sharp and the waveforms appear smoothed or distorted. This suggests that Transformer plays a key role in refining the global waveform morphology.
- (e) **Conv + Transformer:** Without BiLSTM, the model still captures the overall trend, but some waveforms become distorted or irregular. This suggests that BiLSTM helps maintain beat-to-beat consistency and prevents abnormal shape shifts.

This ablation analysis confirms that both the input signal processing and the network architecture design are critical to the performance of CCNN-BP:

- The complex CEEMDAN decomposition is vital for extracting clean cardiac components. Without it (model A), the output suffers from noise and missed beats, especially under motion or respiratory interference.
- Convolutional layers are essential for capturing local waveform features, but using them alone (model B) fails to model temporal patterns, leading to poor shape and alignment.
- The temporal modeling layers—BiLSTM and Transformer—together enable the network to maintain sequence continuity and global timing.
- Removing any one component (models A–E) leads to declines in waveform accuracy, delay estimation, or morphological integrity, confirming that all modules are necessary.
- In contrast, the complete CCNN-BP model consistently produces the best performance across all metrics, especially in waveform alignment and DTW-based shape preservation.

V. CONCLUSION

This study presents an innovative contactless continuous blood pressure monitoring algorithm, CCNN-BP, which integrates Complex CEEMDAN and advanced neural networks. Unlike traditional contactless blood pressure measurement techniques limited to discrete values of systolic and diastolic blood pressure, the proposed method successfully estimates continuous arterial pressure waveforms.

Through Complex CEEMDAN, radar signals reflected from the thoracic surface are effectively decomposed, isolating heartbeat-related IMFs and significantly enhancing the precision of cardiac signal extraction from complex, non-stationary radar data. The extracted IMFs are input into a sophisticated neural network architecture combining one-dimensional convolutional U-Net, bidirectional LSTM, and Transformer encoders, effectively capturing both local morphological features and global temporal dependencies. However, the amplitude of the estimated waveforms remains uncalibrated, requiring an additional amplitude calibration step that is performed individually for each subject in advance. Amplitude calibration uses PTT and uses empirically derived personalised calibration factors to adjust the estimated systolic and diastolic blood pressure values, significantly refining the amplitude accuracy of the waveforms.

Experimental validation conducted using an open-source dataset demonstrated promising accuracy. It is noteworthy that considering the physiological delay of blood flow from the chest (radar measurement location) to the fingertip (TFM measurement location), the designed neural network can effectively infer and compensate for this physiological delay. Logically, it is possible for the neural network to infer this delay from waveform characteristics, as existing studies have extensively modelled the propagation of pulse waves from central (aortic) to peripheral blood vessels [20], [21], [22], [23], [24]. This capability offers significant opportunities for future research. Although certain variability exists among subjects due to differences in physiological conditions and signal quality, the Bland-Altman analysis indicated acceptable levels of systematic bias, suggesting the robustness and potential clinical applicability of the CCNN-BP algorithm. An important limitation lies in the need for individualized amplitude calibration. While the proposed algorithm accurately estimates the morphology of continuous blood pressure waveforms, achieving clinically reliable amplitude still requires subject-specific calibration using personalized parameters. This prerequisite reduces its convenience for real-time or large-scale deployment. In future work, we aim to explore calibration-free solutions by leveraging additional radar-derived physiological indicators (e.g., heart rate variability, respiration patterns) and training large-scale data-driven models to implicitly learn the mapping between radar features and amplitude scaling factors. Such advancements may allow the CCNN-BP system to function in a plug-and-play manner, significantly improving its practicality in daily healthcare settings.

TABLE 6. Characteristics of IMFs.

IMF Index	Spectral Centroid (Hz)	Effective Mean Amp	Description
IMF 1	3.03	1.98	High-frequency noise component (likely instrumentation or quantization noise)
IMF 2	0.50	37.52	Mixed frequency component, possibly overlapping low-amplitude cardiac motion and background interference
IMF 3	1.08	12.25	Potential harmonic or cardiac artifact, weak amplitude makes it less significant
IMF 4	0.88	1.64	Minor component with weak energy, likely residual clutter
IMF 5	0.95	1.93	Similar to IMF 4, weak component possibly arising from filtering artifacts
IMF 6	1.40	13.65	Likely one of the heartbeat-related IMFs (based on frequency and amplitude)
IMF 7	1.13	24.59	Strong heartbeat-related IMF (core contributor to cardiac waveform morphology)
IMF 8	0.37	438.33	Dominant respiration-related component
IMF 9	0.23	403.48	Strong low-frequency respiration or baseline motion
IMF 10	0.47	237.37	Low-frequency motion or residual body movement artifacts
IMF 11	0.40	441.69	Offset/baseline drift component
Residual	0.22	761.74	Overall trend or static bias over the measurement window

It is also important to acknowledge that the dataset used in this study consists solely of healthy individuals. As such, the proposed method has not yet been tested on subjects with hypertension, hypotension, or special physiological conditions such as pregnancy. We recognize this as a current limitation and have outlined it as a key direction for future research. Expanding the dataset to include subjects with more diverse cardiovascular conditions is essential to further evaluate and improve the robustness and generalizability of the proposed algorithm.

Future research should focus on collecting larger and more diverse datasets, enhancing signal quality and robustness, and investigating novel applications leveraging the neural network's capability to compensate for delays between aortic and peripheral blood pressure waveforms. Overall, CCNN-BP emerges as a promising solution for continuous, accurate, and non-intrusive cardiovascular monitoring in daily healthcare scenarios.

APPENDIX INTERPRETATION OF IMFS

This appendix provides a detailed analysis of the intrinsic mode functions (IMFs) obtained from the radar signal segment used in Fig. 3. Each IMF is characterized by its spectral centroid and effective mean amplitude, which respectively represent its frequency-weighted energy distribution and the average amplitude computed while excluding near-zero regions. These indicators help determine whether the IMF

reflects physiological signals such as heartbeat or respiration, or if it corresponds to noise, baseline drift, or other artifacts.

Table 6 below summarizes the characteristics and interpretations for all 12 IMFs and the residual component shown in Fig. 3. It is worth noting that, due to the adaptive and data-driven nature of the CEEMDAN algorithm, the ordering of IMFs does not strictly follow a high-to-low frequency pattern. Therefore, both spectral and amplitude information must be jointly considered when interpreting physiological relevance.

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